Assignment Title
Domain-Specific LLM Assistant using OpenRouter and LangChain

Objective
Build a domain-focused AI assistant using OpenRouter API integrated through LangChain. The assistant must strictly control responses using prompt engineering techniques. Students may choose any domain of their interest (e.g., healthcare, finance, law, education, sports analytics, software engineering, marketing, environmental science, travel advisory, etc.).

Domain Selection

- Choose one specific domain.
- Clearly define the scope of the assistant.
- Clearly define what the assistant will not answer.

Basic Requirements

1. Technical Stack

- Use OpenRouter API for LLM access
- Use LangChain for prompt templates and model orchestration
- Implement a basic LLM chain (agent optional)
- Single-turn interaction is sufficient (multi-turn optional)

2. Prompt Engineering (Mandatory)
   The solution must include:

- Explicit role definition
- Clear domain boundaries
- Structured output format (fixed sections)
- Tone control
- Mandatory disclaimer
- Refusal behavior for out-of-domain queries

3. Domain Control

- The assistant must reject or safely redirect unrelated questions
- The assistant must avoid unsafe or unauthorized advice
- All responses must follow the predefined output structure

4. Testing

- Provide at least 10 test queries
- Include at least 2 out-of-domain queries to demonstrate refusal behavior

Output Criteria

Each response must:

- Follow a consistent structured format
- Stay strictly within the chosen domain
- Maintain consistent professional tone
- Avoid unsupported or fabricated claims

Optional UI Elements (Not Mandatory)

If implementing a UI, you may include:

- Chat-style interface
- Domain selection input
- Temperature control slider
- Conversation history display
- Reset conversation option

Evaluation

- Quality of prompt engineering
- Strict domain adherence
- Output structure consistency
- Safety and refusal handling
- Proper integration of OpenRouter and LangChain

Submission

- Source code
- Prompt template(s)
- Sample outputs
- Short explanation (1–2 pages) describing domain scope, prompt design strategy, and limitations observed